DEEPAK KATUKURI

AI ENGINEER · SQL & AGENTS · MCP



CONTACT

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◆ Potsdam, Germany – open to relocate (Munich)
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TECHNICAL SKILLS

Languages: SQL, PL/SQL, Python, Core Java, C++

AI Agents & MCP: LangGraph, LangChain, tool-calling orchestration, Chroma vector store, Model Context Protocol (MCP) server dev (TypeScript, Apify API)

LLM & RAG: Retrieval-Augmented Generation, hybrid retrieval (BM25 + dense, RRF), cross-encoder reranking, pgvector, sentence-transformers, embeddings, Ollama

Full-Stack & Data: FastAPI, Flask, React, Vite, Tailwind CSS, PostgreSQL, MySQL, SQLAlchemy, Alembic, JWT, Recharts

ML & Testing: Scikit-Learn, NumPy, Pandas, pytest, eval-harness/baseline-pinned regression testing

Cloud & Tools: Docker, Docker Compose, GitHub Actions, CI/CD, Git/GitHub, Linux (Ubuntu)

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present

BTech in Computer Science & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL (Database Management)
Core Java (Software Development)
OneShield (P&C Insurance)

LANGUAGES

English – Fluent
German – A1/A2 (learning)

PROFILE

Data Science MSc candidate with 2.5 years of production software engineering, most of it owning data layers end to end: PL/SQL migration pipelines across customer, policy and claims tables, and pgvector-backed retrieval systems. Built JobPilot, a 16-node LangGraph agentic pipeline with a baseline-pinned eval harness and 238 tests, plus a TypeScript MCP server wrapping the Apify API. Also ships full-stack product — a live React/FastAPI/PostgreSQL health app. Looking for English-speaking working student (Werkstudent) roles in Germany — open to relocating to Munich.

PROJECTS

JobPilot: Multi-Agent Job Search & CV Tailoring Pipeline

Python · LangGraph · Chroma · SQLite · github.com/Deepak589/Job_hunt_Agentic_AI

- Orchestrated a 16-node LangGraph pipeline (requirements → diagnosis → rewrite → fact validation → recruiter/hiring-manager sim → render) gated by a hard validator that blocks any claim without evidence in a Chroma-backed profile store.
- Version-controlled the scoring logic against a baseline-pinned eval harness: a checked-in snapshot of coverage and retrieval metrics that any scoring or threshold change must diff against before it ships, backed by 238 pytest tests across 36 files.
- Sourced real postings for the eval set through a self-built TypeScript MCP server (15 tools, Apify API — actor runs, datasets, key-value store) wired into Claude Code.

RAG Pipeline: Hand-Rolled Retrieval, Validated Against LangChain at Scale

Python · LangChain · BM25 + dense hybrid · pgvector · Ollama · github.com/Deepak589/RAG_pipeline

- Built a hand-rolled hybrid retriever from scratch over 4 iterative stages: BM25 plus a dense bi-encoder, RRF fusion, cross-encoder reranking. Reached 0.833 recall@5 and 0.672 MRR on a 55-question labeled golden set before adopting any framework.
- Rebuilt the same retriever on LangChain using the identical corpus and labels, with the framework as the only changed variable. Caught a silent 0.06-recall regression from LangChain's default BM25 tokenizer and fixed it to 0.854 recall@5, so both implementations matched before scaling up.
- Scaled the corpus ~100x, from 7 to ~740 documents and ~3.5GB by pulling in arXiv, SEC filings, Wikipedia and scanned PDFs through a resumable ingestion pipeline that routes by file type into a pgvector-backed dense store.

RA Nutrition & Lifestyle App: Shipped Full-Stack Product

React · FastAPI · PostgreSQL · Docker · live at ra-project-indol.vercel.app

- Deployed a live full-stack health app as two tiers: a React/Vite/Tailwind frontend on Vercel, and a FastAPI + PostgreSQL backend (4 logical schemas) running as a Docker container on Render, with JWT auth and token refresh.
- Containerised the local stack with Docker Compose: the API, PostgreSQL 16 gated by a pg_isready healthcheck, and an optional pgAdmin profile. Alembic migrations run on container start, so the whole environment comes up with one command.
- Built a rule-based recommendation engine on top of USDA FoodData Central, plus a weekly Recharts dashboard that turns raw meal and symptom logs into specific suggestions for the user.

EXPERIENCE

Software Engineer
Mar 2022 – Sep 2024

ValueMomentum · clients: Hiscox, Philadelphia Insurance

- Migrated full policy, claims, premium and customer history for Philadelphia Insurance's Surety Bonds line from a legacy policy admin system into OneShield PAS. Wrote and validated the PL/SQL migration scripts across customer, policy (PID), claims and premium tables.
- Root-caused a recurring data-integrity defect where claims data for pre-2019 Surety bonds under one bond type loaded empty. The OneShield object copy from the stored object to the loading object failed silently when another bond shared that bond type under a different bond ID. Kept records current with a weekly backfill script while a permanent fix was pending.
- Resolved 100+ production defects across two carrier platforms, Hiscox (P&C insurance) and Philadelphia Insurance (surety bonds). Turned underwriting and claims requirements into technical specs, and ran QA/UAT test cases and CI/CD deployments across the full policy lifecycle.

ML Intern
Jun – Jul 2019

SmartBridge × IBM

- Built a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) and deployed it on IBM Watson ML. Presented the results to non-technical reviewers with Matplotlib and Seaborn charts.